Fundamentals of Statistical Learning

Definitions and results cheatsheet

Chapter 0. Common distributions

Normal distribution

Definition 0.1: Normal distribution

A continuous random variable X has a **Normal** (or **Gaussian**) **distribution** with parameters $(\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$, written $X \sim N(\mu, \sigma^2)$, if its density and CDF are

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \qquad F_X(x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t, \qquad x \in \mathbb{R}.$$

Definition 0.2: Standard Normal

X has a **standard Normal distribution** if $(\mu, \sigma^2) = (0, 1)$. A standard Normal random variable is traditionally denoted by Z, and its density and CDF by

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{z^2}{2}}, \qquad \Phi(z) = \int_{-\infty}^{z} \phi(t)\,\mathrm{d}t, \qquad z \in \mathbb{R}.$$

Proposition 0.1: Standardization

1. Any Normal can be turned into the standard Normal.

   If $X \sim N(\mu, \sigma^2)$, then $Z = \dfrac{X - \mu}{\sigma} \sim N(0, 1)$.

2. The standard Normal can be turned into any Normal.

   Conversely, if $Z \sim N(0, 1)$, then $X = \mu + \sigma Z \sim N(\mu, \sigma^2)$.

3. The quantile of a Normal is obtained by taking the one of the standard Normal and transforming it like any other value, that is, multiplying by σ and adding μ.

   If q_p and z_p are the quantiles of level p of $N(\mu, \sigma^2)$ and of $N(0, 1)$, then $q_p = \mu + z_p\,\sigma$.

Student's t distribution

Definition 0.3: Student's t distribution

X has a **Student's t distribution** with $\nu > 0$ degrees of freedom, written $X \sim t_\nu$, if its density is

$$f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \qquad x \in \mathbb{R}.$$

Definition 0.4: Cauchy distribution

X has a **Cauchy distribution** if its density is

$$f_X(x) = \frac{1}{\pi\,(x^2+1)}, \qquad x \in \mathbb{R},$$

that is, if $X \sim t_1$.

Chapter 1. Learning

The learning problem

Definition 1.1: Likelihood and maximum likelihood

The **likelihood** of θ is the joint density of the observed data, seen as a function of θ. For IID data it is the product

$$L(\theta) = \prod_{i=1}^{n} p_\theta(x_i).$$

The **maximum likelihood estimate** is the value that makes it largest:

$$\hat{\theta} = \arg\max_\theta \; L(\theta).$$

Definition 1.2: Log-likelihood

The **log-likelihood** is the logarithm of the likelihood. For IID data it is the sum

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p_\theta(x_i).$$

Proposition 1.1: Maximum likelihood for a Normal

Let $x_1, \ldots, x_n$ be IID data from $N(\mu, \sigma^2)$. The maximum likelihood estimates of μ and σ^2 are

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Algorithm 1.1: Learning a model from data

1. **Data.** Collect IID data $\mathcal{D}_n = (x_1, \ldots, x_n)$.
2. **Model.** Choose a family of distributions p_θ, with parameter θ.
3. **Likelihood.** Write the likelihood of the data, $L(\theta) = \prod_{i=1}^{n} p_\theta(x_i)$.
4. **Maximum likelihood.** Choose $\hat{\theta} = \arg\max_\theta L(\theta)$, or equivalently $\arg\max_\theta \ell(\theta)$.

Hidden variables and mixtures

Definition 1.3: Simplex

The **simplex** of dimension $K-1$ is the set

$$\Delta^{K-1} = \left\{ (p_1, \ldots, p_K) \in \mathbb{R}^K \;\middle|\; p_i \geq 0,\ \sum_{i=1}^{K} p_i = 1 \right\}.$$

Definition 1.4: Finite mixture model

A **finite mixture model** with K **components** has density (or mass function)

$$f_{\text{obs}}(o) = \sum_{j=1}^{K} f_j(o)\, \pi_j,$$

where $f_1, \ldots, f_K$ are the distributions of the K groups and $(\pi_1, \ldots, \pi_K) \in \Delta^{K-1}$ are their weights.

Definition 1.5: Gaussian mixture model

A **Gaussian mixture model** (GMM) with K components has density

$$f_{\text{obs}}(o) = \sum_{j=1}^{K} \pi_j\, N(o \mid \mu_j, \sigma_j^2),$$

with parameters $\theta = \{\pi_j,\, \mu_j,\, \sigma_j^2\}_{j=1}^{K}$, where $(\pi_1, \ldots, \pi_K) \in \Delta^{K-1}$.

Proposition 1.2: Maximum likelihood for a GMM with labels

Let $o_1, \ldots, o_n$ be IID data from a GMM with K components, and suppose that the group h_i of every observation is known. Let n_j be the number of observations in group j. Then the maximum likelihood estimates are

$$\hat{\pi}_j = \frac{n_j}{n}, \qquad \hat{\mu}_j = \frac{1}{n_j} \sum_{i:\, h_i = j} o_i, \qquad \hat{\sigma}_j^2 = \frac{1}{n_j} \sum_{i:\, h_i = j} (o_i - \hat{\mu}_j)^2, \qquad j = 1, \ldots, K.$$

Chapter 2. Expected value

Why summarise a distribution?

Definition 2.1: Expected value: the intuitive idea

The **expectation**, **expected value** or **mean** $\mathbb{E}(X)$ is the weighted average of the possible values of X, with weights equal to their probabilities:

$$\mathbb{E}(X) = \sum_{x} x \cdot P(X = x).$$

The definition

Definition 2.2: Expected value

The **expected value** (expectation, mean, first moment) of $X \sim F_X$ is

$$\mathbb{E}(X) = \begin{cases} \displaystyle\sum_x x\, p_X(x) & \text{if } X \text{ is discrete with PMF } p_X, \\ \displaystyle\int_{\mathbb{R}} x\, f_X(x)\, \mathrm{d}x & \text{if } X \text{ is continuous with PDF } f_X, \end{cases}$$

provided the sum or the integral is well defined, that is $\sum_x |x|\, p_X(x) < \infty$ or $\int_{\mathbb{R}} |x|\, f_X(x)\, \mathrm{d}x < \infty$.

Both cases can be written at once as

$$\mathbb{E}(X) = \int_{\mathbb{R}} x\, \mathrm{d}F_X(x),$$

where $\mathrm{d}F_X(x)$ stands for $f_X(x)\, \mathrm{d}x$ in the continuous case, and the integral becomes the sum $\sum_x x\, p_X(x)$ in the discrete case.

Definition 2.3: Moments

For $k = 1, 2, 3, \ldots$, the **k-th moment** of X is the expected value of its k-th power, $\mathbb{E}(X^k)$, provided it exists. The expected value $\mathbb{E}(X)$ is the first moment.

Property 0: distribution specific

Proposition 2.1: Distribution specific

$\mathbb{E}(X)$ depends only on the distribution of X: two random variables with the same distribution have the same expected value.

Property 1: centre of mass

Proposition 2.2: Symmetric distributions

Since $\mathbb{E}(X)$ is a centre of mass, if the distribution of X is symmetric about a point x_0 *and* $\mathbb{E}(X)$ exists, then $\mathbb{E}(X) = x_0$.

Proposition 2.3: Lack of robustness

The expected value can be affected greatly by a very small change in the probability assigned to a large value of x. Moving an amount ε of probability from a value x_j to a point M shifts $\mathbb{E}(X)$ by $\varepsilon\,(M - x_j)$: however small ε is, choosing M far enough moves $\mathbb{E}(X)$ arbitrarily far away.

Property 3: the law of the lazy statistician

Theorem 2.1: Law of the lazy statistician

Let $X \sim F_X$ and $Y = r(X)$ for a deterministic function r. Then

$$\mathbb{E}(Y) = \mathbb{E}\big(r(X)\big) = \begin{cases} \displaystyle\sum_x r(x)\, p_X(x) & \text{if } X \text{ is discrete,} \\ \displaystyle\int_{\mathbb{R}} r(x)\, f_X(x)\, \mathrm{d}x & \text{if } X \text{ is continuous.} \end{cases}$$

Corollary 2.1: Probabilities are expected values

For any event A,

$$\mathbb{E}\big(\mathbb{I}_A(X)\big) = \int_{\mathbb{R}} \mathbb{I}_A(x)\, f_X(x)\, \mathrm{d}x = \int_A f_X(x)\, \mathrm{d}x = P(X \in A).$$

Property 4: Jensen inequality

Proposition 2.4: Linear functions

If $r(x) = \alpha + \beta x$ for constants α and β, then

$$\mathbb{E}\big(r(X)\big) = \mathbb{E}(\alpha + \beta X) = \alpha + \beta\, \mathbb{E}(X) = r\big(\mathbb{E}(X)\big).$$

Proposition 2.5: Convexity and second derivative

If r is twice differentiable, then r is convex on $S \subseteq \mathbb{R}$ if and only if $r''(x) \geq 0$ for all $x \in S$.

Definition 2.4: Convex function

A function $r : \mathbb{R} \to \mathbb{R}$ is **convex** on $S \subseteq \mathbb{R}$ if, for any $x_1, x_2 \in S$ and $t \in [0, 1]$,

$$r\big(t\, x_1 + (1 - t)\, x_2\big) \leq t\, r(x_1) + (1 - t)\, r(x_2).$$

It is **concave** if $-r$ is convex.

Theorem 2.2: Jensen's inequality

Let $X \sim F_X$ and let r be a convex function, with $\mathbb{E}(X)$ and $\mathbb{E}\big(r(X)\big)$ existing. Then

$$r\big(\mathbb{E}(X)\big) \leq \mathbb{E}\big(r(X)\big).$$

If r is concave, the inequality is reversed.

Definition 2.5: ELBO

Let obs be the observed variable, hid the hidden one, and q any distribution of hid. The **evidence lower bound** (ELBO) is

$$F(q,\theta) = \mathbb{E}_q\left[\log \frac{p_\theta(\text{obs},\text{hid})}{q(\text{hid})}\right].$$

Evidence refers to $\log p_\theta(\text{obs})$, the log-likelihood of what we observe, and *lower bound* to the fact that $F(q,\theta) \leq \log p_\theta(\text{obs})$ for every q and every θ.

Definition 2.6: Information gap

The **information gap** is the difference between the log-likelihood and the ELBO,

$$\log p_\theta(\text{obs}) - F(q,\theta).$$

Definition 2.7: Kullback–Leibler divergence

The **Kullback–Leibler divergence** (KL) of q from p is

$$\text{KL}(q \mid p) = \int q(x) \log \frac{q(x)}{p(x)}\,\mathrm{d}x.$$

It is not a distance, and $\text{KL}(q \mid p) = 0$ if and only if $q = p$.

Algorithm 2.1: EM algorithm

Start from a parameter $\theta^{(0)}$. For $t = 1, 2, \ldots$ repeat:

1. **E-step.** Optimise over all distributions q, with θ fixed:
$$q^{(t)} = \arg\max_q F\big(q, \theta^{(t-1)}\big).$$
2. **M-step.** Optimise over all parameters θ, with q fixed:
$$\theta^{(t)} = \arg\max_\theta F\big(q^{(t)}, \theta\big).$$

Definition 2.8: Responsibilities

The **responsibility** of group j for the observation o_i is the posterior probability of its hidden variable,

$$r_j^{(i)} = p_\theta(\text{hid}_i = j \mid o_i), \qquad i = 1,\ldots,n, \quad j = 1,\ldots,K.$$

Property 5: variational characterisations

Definition 2.9: Loss function

A **loss function** $L(X, d)$ measures how bad the guess d is when the value turns out to be X. The most used are

$$\underbrace{(X-d)^2}_{\text{square loss}}, \qquad \underbrace{|X-d|}_{\text{abs loss}}, \qquad \underbrace{\mathbb{I}(X \neq d)}_{\text{0-1 loss}}.$$

Definition 2.10: Risk function

The **risk function** is the expected loss,

$$R(d) = \mathbb{E}\, L(X, d) = \int_{\mathbb{R}} L(x, d)\, \mathrm{d}F_X(x),$$

where the integral comes from the law of the lazy statistician. The best guess minimises the risk:

$$d^* = \arg\min_d \ \mathbb{E}\, L(X, d).$$

Definition 2.11: Mean squared error

The risk of the square loss,

$$R(d) = \mathbb{E}\, L(X, d) = \mathbb{E}(X-d)^2,$$

is called the **mean squared error** (MSE) of the guess d.

Proposition 2.6: Variational characterisation of $\mathbb{E}(X)$

Let X be a random variable with $\mathbb{E}(X^2) < \infty$, and let $\mu = \mathbb{E}(X)$. Then, for every number d,

$$\mathbb{E}(X-\mu)^2 \leq \mathbb{E}(X-d)^2,$$

with equality if and only if $d = \mu$. So $d^* = \mu$.

Definition 2.12: Mean absolute error

The risk of the abs loss,

$$R(d) = \mathbb{E}\, L(X, d) = \mathbb{E}\, |X-d|,$$

is called the **mean absolute error** (MAE) of the guess d.

Proposition 2.7: Variational characterisation of the median

Let X be a random variable with $\mathbb{E}\,|X| < \infty$, and let m be a median of X. Then, for every number d,

$$\mathbb{E}\, |X-m| \leq \mathbb{E}\, |X-d|,$$

with equality if and only if d is also a median of X. So $d^* = m$.

Variance and standard deviation

Definition 2.13: Variance

Let $X \sim F_X$ have expected value $\mu = \mathbb{E}(X) < \infty$. The **variance** of X is the expected squared gap between X and μ,

$$\mathrm{Var}(X) = \mathbb{E}\big((X-\mu)^2\big) = \int_{\mathbb{R}} (x-\mu)^2 \, \mathrm{d}F_X(x).$$

Definition 2.14: Standard deviation

The **standard deviation** of X is

$$\mathrm{sd}(X) = \sigma = \sqrt{\mathrm{Var}(X)}.$$

Definition 2.15: Median absolute deviation

The **median absolute deviation** of X is

$$\mathrm{MAD}(X) = \mathrm{Median}\,\big|X - \mathrm{Median}(X)\big|.$$

Proposition 2.8: Existence of the variance

For every random variable X, $\mathrm{Var}(X) \geq 0$. If X is bounded, then $\mathrm{Var}(X)$ exists and is finite.

Proposition 2.9: Degeneracy

Let X have variance σ^2. Then

$$\sigma^2 = 0 \quad \Longleftrightarrow \quad X = \mathbb{E}(X) \text{ almost surely.}$$

Proposition 2.10: Variance from the moments

For every random variable X with finite variance,

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2.$$

Proposition 2.11: Existence of lower moments

If the k-th moment of X exists and $j < k$, then the j-th moment of X exists.

Going multivariate

Definition 2.16: Expected value of a random vector

The expected value of a random vector $\underline{X} = (X_1, \ldots, X_p)^\mathsf{T}$ is the vector of the expected values of its components,

$$\mathbb{E}(\underline{X}) = \begin{pmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_p) \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix} = \underline{\mu}.$$

Each $\mu_j = \mathbb{E}(X_j)$ is the **marginal expectation** of X_j.

Proposition 2.12: Linear combinations

Let $\underline{X}$ be a random vector with $\underline{\mu} = \mathbb{E}(\underline{X})$, let $\underline{\alpha} \in \mathbb{R}^p$ be a vector of constants, and let $Y = \underline{\alpha}^\mathsf{T}\underline{X}$. Then

$$\mathbb{E}(Y) = \underline{\alpha}^\mathsf{T}\underline{\mu} = \langle \underline{\alpha}, \underline{\mu} \rangle.$$

Proposition 2.13: Expected value of a product

Let $\underline{X} = (X_1, \ldots, X_p)^\mathsf{T}$ have independent components with $\mu_j = \mathbb{E}(X_j)$, and let $Y = \prod_{j=1}^p X_j$. Then

$$\mathbb{E}(Y) = \prod_{j=1}^{p} \mu_j.$$

Definition 2.17: Covariance matrix

The **covariance matrix** of $\underline{X}$, with $\underline{\mu} = \mathbb{E}(\underline{X})$, is the $p \times p$ matrix

$$\mathrm{Cov}(\underline{X}) = \mathbb{E}\big(\underline{X}\,\underline{X}^\mathsf{T}\big) - \underline{\mu}\,\underline{\mu}^\mathsf{T},$$

where

$$\underline{X}\,\underline{X}^\mathsf{T} = \begin{pmatrix} X_1X_1 & X_1X_2 & \cdots & X_1X_p \\ X_2X_1 & X_2X_2 & \cdots & X_2X_p \\ \vdots & \vdots & \ddots & \vdots \\ X_pX_1 & X_pX_2 & \cdots & X_pX_p \end{pmatrix}, \qquad \underline{\mu}\,\underline{\mu}^\mathsf{T} = \begin{pmatrix} \mu_1\mu_1 & \mu_1\mu_2 & \cdots & \mu_1\mu_p \\ \mu_2\mu_1 & \mu_2\mu_2 & \cdots & \mu_2\mu_p \\ \vdots & \vdots & \ddots & \vdots \\ \mu_p\mu_1 & \mu_p\mu_2 & \cdots & \mu_p\mu_p \end{pmatrix}.$$

Its entry (i, j) is the **covariance** of X_i and X_j,

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}(X_iX_j) - \mathbb{E}(X_i)\,\mathbb{E}(X_j) = \mathbb{E}\big((X_i - \mu_i)(X_j - \mu_j)\big),$$

where $\mathbb{E}(X_iX_j)$ is the **cross-moment** of X_i and X_j.

Proposition 2.14: Variance of a linear combination

Let $\underline{X}$ be a random vector, let $\underline{\alpha} \in \mathbb{R}^p$ be a vector of constants, and let $Y = \underline{\alpha}^\mathsf{T} \underline{X}$. Then

$$\operatorname{Var}(Y) = \underline{\alpha}^\mathsf{T} \operatorname{Cov}(\underline{X})\, \underline{\alpha} = \sum_{j=1}^{p} \alpha_j^2 \operatorname{Var}(X_j) + 2 \sum_{i<j} \alpha_i \alpha_j \operatorname{Cov}(X_i, X_j).$$

Corollary 2.2: Variance of a sum of independent variables

If the components of $\underline{X}$ are independent, then

$$\operatorname{Var}\Big(\sum_{j=1}^{p} \alpha_j X_j \Big) = \sum_{j=1}^{p} \alpha_j^2 \operatorname{Var}(X_j).$$

In particular, with all $\alpha_j = 1$, the variance of a sum is the sum of the variances.

Conditioning

Definition 2.18: Conditional expectation and variance given $X = x$

The **conditional expectation** and the **conditional variance** of Y given $X = x$ are

$$\mathbb{E}(Y \,|\, X = x) = \int_{\mathbb{R}} y \, \mathrm{d}F_{Y|X}(y \,|\, x),$$
$$\operatorname{Var}(Y \,|\, X = x) = \int_{\mathbb{R}} \big(y - \mathbb{E}(Y \,|\, X = x)\big)^2 \, \mathrm{d}F_{Y|X}(y \,|\, x).$$

Both are numbers, which depend on x. Swapping the roles of X and Y gives $\mathbb{E}(X \,|\, Y = y)$ and $\operatorname{Var}(X \,|\, Y = y)$.

Definition 2.19: Conditional expectation

Let $r(x) = \mathbb{E}(Y \,|\, X = x)$. The **conditional expectation** of Y given X is the random variable

$$\mathbb{E}(Y \,|\, X) = r(X).$$

Theorem 2.3: Law of total expectation

For random variables X and Y,

$$\mathbb{E}_Y(Y) = \mathbb{E}_X\big(\mathbb{E}_{Y|X}(Y \,|\, X)\big).$$

Corollary 2.3: Law of total probability

For any events A and B,

$$P(B) = P(B \,|\, A)\, P(A) + P(B \,|\, A^c)\, P(A^c).$$